

Houxiang Ji

CONTACT INFORMATION	213 Coordinated Science Lab 1308 W Main St., Urbana, IL 61801	Website: https://hxji.github.io/ Email: hj14@illinois.edu
RESEARCH INTERESTS	Memory Tiering System, Compute eXpress Link (CXL), System for ML	
EDUCATION	University of Illinois Urbana-Champaign Ph.D. Candidate, Computer Science Advisor: Prof. Nam Sung Kim Shanghai Jiao Tong University B.Eng., Computer Science and Engineering, July 2018 B.Sc., Finance and Economics, July 2018	
HONORS AND AWARDS	• Best Demo Award, SRC/DARPA PRISM Center Annual Review, 2023 • UIUC CS PhD Fellowship, UIUC, 2023 • Travel Grants: USENIX OSDI/ATC 2023, IEEE ISCA 2019 • Ray Ozzie Computer Science Fellowship, UIUC, 2018 • Zhiyuan Honor Degree of B.Sc. in Computer Science, SJTU, 2018 • Outstanding Graduates, SJTU, 2018 • Zhiyan Honor Scholarship, SJTU, 2015-2017 • Excellent Undergraduate Scholarship, SJTU, 2015-2017	
RESEARCH EXPERIENCE	CXL Analysis and Utilization in Datacenter Supervised by Prof. Nam Sung Kim • explore CXL Type-2 device to boost hyperscaler application performance [P1] • CXL device characterization with genius system [C11, P1] Datacenter Memory Tax Reduction Supervised by Prof. Nam Sung Kim • leverage SmartNIC to reduce kernel features cost on CPU [C10] • kernel feature acceleration with near-memory processing unit (AxDIMM) • explore on-chip accelerator (e.g. QAT, DSA) to reduce datacenter tax Deep Learning Model Acceleration on CPU Supervised by Prof. Josep Torrellas • sw/hw co-design to optimize aggregation/update phase in GNN [C8] • leverage sparsity in DNN for faster training and inference [C5, C6] • demystify graph neural networks in recommender systems [master thesis]	
PROFESSIONAL EXPERIENCE	Capacity Engineering and Analysis, Meta Visiting Researcher <i>Explored CXL devices in collaboration with Meta Infrastructure, focusing on their integration and performance optimization within cutting-edge computing environments.</i> CPU Design, Luminous Computing Research Intern <i>Design and simulate a RISC-V-based AI CPU architecture optimized for high-bandwidth photonic data transmission.</i>	

Security and Privacy Research, Intel Lab

Research Intern

May 2020 - Aug 2020

Implement and evaluate hardware/software co-design defense schemes against speculative execution attacks using a commercial CPU simulator.

CONFERENCE &
JOURNAL
PUBLICATIONS

[C13] Jinghan Huang, Jiaqi Lou, Srikar Vanavasam, Xinhao Kong, **Houxiang Ji**, Ipoom Jeong, Eun Kyung Lee, Danyang Zhuo, Nam Sung Kim. *HAL: Hardware-assisted Load Balancing for Energy-efficient SNIC-Host Cooperative Computing*, **IEEE/ACM ISCA 2024, to appear**

[C12] Chihun Song, Michael Jaemin Kim, Tianchen Wang, **Houxiang Ji**, Jinghan Huang, Ipoom Jeong, Jaehyun Park, Hwayong Nam, Minbok Wi, Jung Ho Ahn, Nam Sung Kim. *TAROT: A CXL SmartNIC-Based Defense Against Multi-bit Errors by Row Hammer Attacks*, **ACM ASPLOS 2024, to appear**

[C11] Yan Sun, Yifan Yuan, Zeduo Yu, Chihun Song, Reese Kuper, Jinghan Huang, **Houxiang Ji**, Siddharth Agarwal, Jiaqi Lou, Ipoom Jeong, Ren Wang, Jung Ho Ahn, Tianyin Xu, Nam Sung Kim. *Demystifying CXL Memory with Genuine CXL-Ready Systems and Devices*, **IEEE/ACM MICRO 2023**

[C10] **Houxiang Ji**, Yan Sun, Mark Mansi, Yifan Yuan, Jinghan Huang, Reese Kuper, Michael M. Swift, Nam Sung Kim. *STYX: Exploiting SmartNIC Capability to Reduce Datacenter Memory Tax*, **USENIX ATC 2023**

[C9] Lihui Liu, **Houxiang Ji**, Jiejun Xu, and Hanghang Tong. *Comparative Reasoning for Knowledge Graph Fact Checking*, **IEEE BigData 2022**

[C8] Zhangxiaowen Gong, **Houxiang Ji**, Yao Yao, Christopher Fletcher, Christopher Hughes, Josep Torrellas. *Optimizing Graph Neural Networks on CPUs via Cooperative Software-Hardware Techniques*, **IEEE/ACM ISCA 2022**

[C7] Zirui Neil Zhao, **Houxiang Ji**, Adam Morrison, Darko Marinov, Josep Torrellas. *Pinned Loads: Taming Speculative Loads in Secure Processors*, **ACM ASPLOS 2022**

[C6] Zhangxiaowen Gong, **Houxiang Ji**, Christopher Fletcher, Christopher Hughes, Josep Torrellas. *SparseTrain: Leveraging Dynamic Sparsity in Software for Training DNNs on General-Purpose SIMD Processors*, **PACT 2020 [Open Source]**

[C5] Zhangxiaowen Gong, **Houxiang Ji**, Christopher Fletcher, Christopher Hughes, Sara Baghsorkhi, Josep Torrellas. *SAVE: Sparsity-Aware Vector Engine for Accelerating DNN Training and Inference on CPUs*, **IEEE/ACM MICRO 2020**

[C4] Zirui Zhao, **Houxiang Ji**, Mengjia Yan, Jiyong Yu, Christopher W. Fletcher, Adam Morrison, Darko Marinov, Josep Torrellas. *Speculation Invariance (InvarSpec): Faster Safe Execution Through Program Analysis*, **IEEE/ACM MICRO 2020**

[C3] **Houxiang Ji**, Li Jiang, Tianjian Li, Naifeng Jing, Jing Ke, Xiaoyao Liang. *HUBPA: High Utilization Bidirectional Pipeline Architecture for Neuromorphic Computing*, **ASP-DAC 2019**

[C2] **Houxiang Ji**, Linghao Song, Li Jiang, Hai(Halen) Li, Yiran Chen. *ReCom: An efficient resistive accelerator for compressed deep neural networks*, **IEEE DATE 2018**

[C1] Luyu Li, **Houxiang Ji**, Chentao Wu, Jie Li, Minyi Guo. *Favorable Block First: A Comprehensive Cache Scheme to Accelerate Partial Stripe Recovery of Triple Disk Failure Tolerant Array*, **ICPP 2017**

INVITED TALKS

Unveiling CXL Type 2 Device: From Characterization to Deployment

- OCP composable memory system group talk, Mar. 2024

STYX: Exploiting SmartNIC Capability to Reduce Datacenter Memory Tax

- USENIX ATC 2023

Improve System Performance by Offloading Memory-Intensive Kernel Features to CXL Type-2 Device

- OCP global summit 2023
- OCP composable memory system group talk, Oct. 2023

PROFESSIONAL
SERVICE

Reviewer

- IEEE Transactions on Emerging Topics in Computing, 2024
- ACM Journal on Emerging Technologies in Computing Systems, 2024
- IEEE International Conference on Artificial Intelligence Circuits and Systems, 2021,2023
- IEEE International Conf. on Networks, Communication and Information Tech, 2022

Artifact Evaluation Committee

- 2024: OSDI, ATC

Community Service

- Mentor, Promoting Undergraduate Research in Engineering (PURE), UIUC, 2023
- Mentor, Undergraduate Researcher in [FAST](#), UIUC, 2023

TEACHING
EXPERIENCE
REFERENCES

CS233 Computer Architecture, Teaching Assistant, UIUC Spring 2024

References available upon request.